mahmoud mossad

AI Automation & Backend Engineer

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LinkedIn • GitHub

Profile

Backend developer with experience building APIs and automation workflows. Worked with Node.js/Express on backend systems, and used n8n to automate business processes and connect tools via webhooks. Also built AI agent pipelines that integrate with external services. I focus on writing clean, maintainable code and designing systems that are reliable in production. Comfortable with Python, Redis, and SQL.

Skills

Programming Languages: TypeScript, JavaScript (ES6+), Python, HTML/CSS

Automation & AI: n8n, Workflow Orchestration, LLM Integration (OpenAI/Gemini), Multi-Agent Systems, Prompt Engineering, Webhooks

Backend Frameworks & APIs: Node.js, Express.js, Nest.js, FastAPI, RESTful API Design, OAuth 2.0, JWT, Authentication & Authorization

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis (Session Management, Chat Memory)

DevOps & Tools: Git, GitHub, Postman, Linux, Docker (Basics)

CS Fundamentals: Data Structures, Algorithms, Object-Oriented Programming (OOP), Design Patterns, SQL Database Design

Web Scraping & Data Extraction: Selenium, BeautifulSoup, Structured Data Extraction

Experience

Automation Engineer at Tawiq (Tech Startup), Port Said, Egypt **April 2026 - Present**

Tawiq – Tech startup specializing in workflow automation and AI Agents using LLMs, serving the Saudi and MENA markets.

- Set up automatic PDF report delivery via Telegram and Gmail, which saved the team 90% of the time they used to spend on manual reporting.
- Engineered an n8n workflow that handles 50+ rider complaints daily via Telegram, supporting text and voice with 95% classification accuracy.
- Added automatic language detection to the platform, supporting 4+ languages and making it easier for users to communicate across teams.
- Designed n8n workflows to scale automatically with rider spikes, keeping extra infrastructure costs at $0.

Projects

WhatsApp AI Agent with Redis Memory • GitHub **June 2026 - July 2026**

Designed a conversational customer support agent for WhatsApp with Redis-backed long-term memory, achieving 98% session retention.

- Deployed a production-ready WhatsApp support bot using OpenRouter (Gemma) with Redis Chat Memory.
- Developed conditional routing logic to detect first-time vs. returning customers.

AutoFlow - Order Intake Workflow • GitHub **March 2026 - March 2026**

Orchestrated a lightweight n8n workflow that captures orders from WhatsApp/Telegram, processing 20+ orders per day with 90% accuracy.

- Built n8n workflows triggered by webhooks to instantly validate incoming orders.
- Integrated Gemini AI for Arabic text extraction (90%+ accuracy).

AI Chatbot API • Github **October 2025 - November 2025**

Scalable AI chatbot backend supporting 3 AI models with context-aware conversations, image processing, dual authentication (JWT + Cookies), rate limiting, and SQLite-based history storage.

- Built an AI backend that injects user context directly into prompts, handling 1,000+ concurrent connections with <200ms response time.

Islamatic-Server • **GitHub** **March 2025 - May 2025**

Built a Node.js/Express backend with Google OAuth for secure audio uploads, with a flexible architecture for future media processing features.

- Implemented secure authentication and file management with email notifications, supporting large audio file uploads and conducting security testing.

Education

Bachelor of Science (BSc) at Computers and Information , Suez Canal University, Ismailia, Egypt **November 2023 - Present**

Grade: 3.1

Internships

Gen AI at Information Technology Institute (ITI), Ismailia, Al Isma'iliyah, Egypt **August 2025 - September 2025**

- Completed 120+ hours of training in Prompt Engineering, Python for AI, and data cleaning.
- Collaborated with a 5-member team to build an AI chatbot integrating AI APIs with backend workflows.

Languages

Arabic - Native

English - Conversational